

Technology Stack

Requirement Analysis Phase — Architecture & Stack

Project Name	Automated Network Request Management in ServiceNow
Phase	Phase 2: Requirement Analysis
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Date	29 June 2026
Platform / Domain	ServiceNow / IT Service Management (ITSM)
Academic Year	2025 – 2026

1. Technical Architecture Summary

The Automated Network Request Management solution utilizes ServiceNow's multi-layered enterprise cloud architecture. The system is organized into four primary layers, each responsible for a distinct aspect of the request lifecycle:

Presentation Layer

Provides structured form interfaces to end users through the unified ServiceNow Service Portal and native internal ITSM workspaces for engineers and approvers.

Application / Business Logic Layer

Enforces data security and validates complex network configuration inputs using Client Scripts and UI Policies. Manages role-based access through Access Control Lists (ACLs).

Automation Layer

Manages end-to-end request state routing using Flow Designer. Orchestrates multi-stage approval chains, conditional branching, SLA clock management, and event-driven email notification dispatch.

Storage Layer

Securely houses standard ITSM tables (sc_req_item, sysapproval_approver, sc_task) and the custom u_network_database table in the ServiceNow cloud database with full audit trail support.

2. Components & Technologies

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1	User Interface	Dedicated requester portal and centralized workspaces for network operators.	ServiceNow Service Portal, Service Catalog UI, Native Forms & Lists
2	Application Logic	Dynamic conditional form behaviors, field hiding, regex parameter validation for IP/VLAN/port inputs.	UI Policies, Client Scripts, Data Policies
3	Workflow Automation	Multi-stage approval loops, sequential task distributions, and live operational escalations.	Flow Designer, Core Approval Engine, Service Catalog Workflow
4	Notifications	Automated event-driven email dispatch for all lifecycle stage transitions.	ServiceNow Notification Engine, Email Templates
5	Data Storage	Standard ITSM tables and custom u_network_database for operational logging.	ServiceNow Cloud Database (sc_req_item, sc_task, sysapproval_approver, u_network_database)
6	Security	Role-based data access control and field-level permission enforcement.	Access Control Lists (ACLs), sys_user Roles
7	SLA Management	Real-time SLA tracking with automated escalation at configurable breach thresholds.	ServiceNow SLA Engine, Escalation Rules
8	Deployment	Configuration packaging and environment migration without manual re-entry.	ServiceNow Update Sets
9	Reporting	Request volume monitoring, SLA performance dashboards, and productivity metrics.	ServiceNow Reporting & Performance Analytics

3. Technology Stack Justification

ServiceNow Platform

Chosen as the delivery platform due to its native ITSM modules, no-code/low-code automation capabilities, enterprise-grade security model, and proven scalability in large IT operations environments.

Flow Designer

Selected over legacy Workflow Editor due to its visual, action-based design interface, better support for conditional branching, and native integration with the SLA and notification engines.

Service Catalog

Used as the structured request intake mechanism because it provides consumer-grade usability for end users while enforcing backend data validation and workflow triggers automatically.

Client Scripts & UI Policies

Applied for real-time form-level validation and dynamic field visibility, ensuring data quality at the point of entry before submission rather than during back-end processing.